

Project title

User profile insights using API

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April 2026

1.introduction

الهدف من المشروع: استخدام API وتحليل البيانات باستخدام Python & Pandas:

2.Data collection using API

```
[6]: for user in user:
      print(user["name"] ["first"])

Lance
Joann
Parcício
Derrick
Constance
Miguel
Alice
Katharina
Rishaan
Nanna

[1]: import requests
      import pandas as pd

[4]: url = "https://randomuser.me/api/?results=10"
      response = requests.get(url)
      data = response.json()["results"]
      users = []
      for user in data:
          users.append({
              "First Name": user["name"] ["first"],
              "Last Name": user["name"] ["last"],
              "Gender": user["gender"],
              "Age": user["dob"] ["age"],
              "Country": user["location"] ["country"],
              "Email": user["email"]
          })
      df = pd.DataFrame(users)
      df
```

Figure 1

```
[7]: import requests
      import pandas as pd

[2]: import requests
      url = "http://randomuser.me/api/?results=10"
      response = requests.get(url)
      data = response.json()
      data

[2]: {'results': [{'gender': 'male',
                  'name': {'title': 'Mr', 'first': 'Lance', 'last': 'Palmer'},
                  'location': {'street': {'number': 8716, 'name': 'Mill Lane'},
                              'city': 'Wexford',
                              'state': 'Meath',
                              'country': 'Ireland',
                              'postcode': 13281,
                              'coordinates': {'latitude': '-4.0996', 'longitude': '-52.3854'},
                              'timezone': {'offset': '-1:00',
                                           'description': 'Azores, Cape Verde Islands'}},
                  'email': 'lance.palmer@example.com',
                  'login': {'uuid': '9b6d6fa0-1014-426d-80f1-0ca2daf99ed9',
                           'username': 'whitebutterfly153',
                           'password': '1995',
                           'salt': 'hafrjxYX',
                           'md5': '212548618a06a26c17199afe065af808',
                           'sha1': 'a7ef863a6c12cb47933892400d7aef99864dc5c2',
                           'sha256': '407338e665c5c7f38e7a7cef397a66b7580693a9037f4b986c53575c5bb08a69'},
                  'dob': {'date': '1990-05-28T03:25:04.953Z', 'age': 35},
                  'registered': {'date': '2015-03-21T02:55:37.882Z', 'age': 11},
                  'phone': '061-696-5985',
                  'cell': '081-355-4963',
                  'id': {'name': 'PPS', 'value': '4316314T'},
                  'picture': {'large': 'https://randomuser.me/api/portraits/men/17.jpg',
                              'medium': 'https://randomuser.me/api/portraits/med/men/17.jpg',
                              'thumbnail': 'https://randomuser.me/api/portraits/thumb/men/17.jpg'},
                  'nat': 'IE'},
        {'gender': 'female',
         'name': {'title': 'Ms', 'first': 'Joann', 'last': 'Phillips'},
```

Figure2

3.Data processing with Pandas

```
[4]:
```

	First Name	Last Name	Gender	Age	Country	Email
0	Sayali	Patil	female	52	India	sayali.patil@example.com
1	Silas	Johansen	male	79	Denmark	silas.johansen@example.com
2	Alma	Nielsen	female	67	Denmark	alma.nielsen@example.com
3	Hilda	Phillips	female	45	Australia	hilda.phillips@example.com
4	Jordan	Nguyen	male	42	United Kingdom	jordan.nguyen@example.com
5	Marinelza	Lopes	female	34	Brazil	marinelza.lopes@example.com
6	Zachary	Macdonald	male	36	Canada	zachary.macdonald@example.com
7	Gordon	Davis	male	67	Australia	gordon.davis@example.com
8	Leticia	Sotelo	female	40	Mexico	leticia.sotelo@example.com
9	Emilie	Mitchell	female	62	Canada	emilie.mitchell@example.com

```
[6]: df["Age"].mean()
```

```
[6]: np.float64(52.4)
```

```
[7]: df["Gender"].value_counts()
```

```
[7]: Gender
female    6
male      4
Name: count, dtype: int64
```

Figure 3

4.Statistical Analysis

```
[6]: df["Age"].mean()
[6]: np.float64(52.4)
[7]: df["Gender"].value_counts()
[7]: Gender
female    6
male      4
Name: count, dtype: int64
[8]: df["Country"].value_counts()
[8]: Country
Denmark      2
Australia    2
Canada       2
India        1
United Kingdom 1
Brazil       1
Mexico       1
Name: count, dtype: int64
```

```
[18]: len(df)
```

```
[18]: 10
```

Figure 4

5.Data Visualization

```
[11]: df["Gender"].value_counts().plot(kind="bar")
```

```
[11]: <Axes: xlabel='Gender'>
```

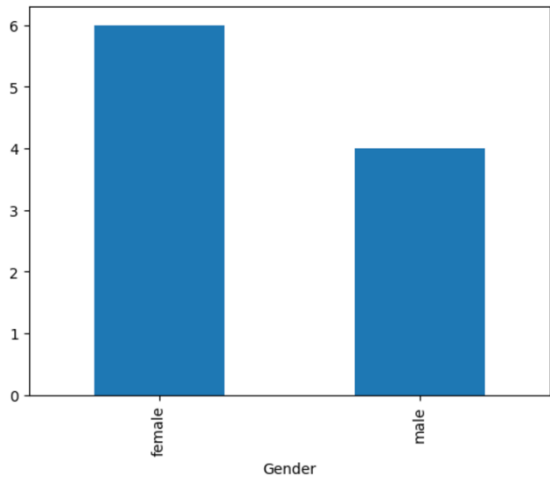
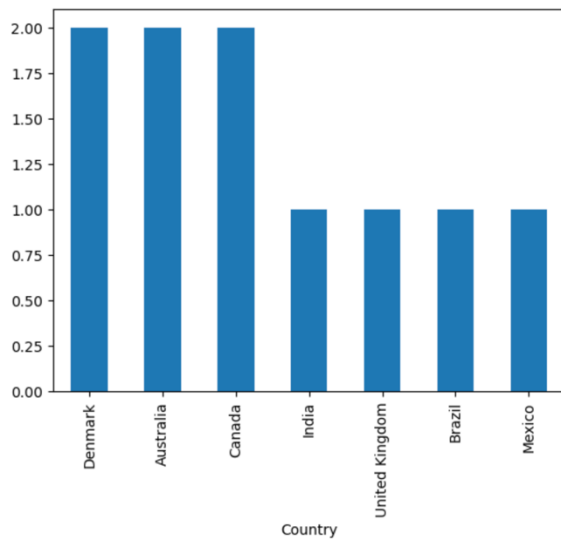


Figure 5

```
[17]: df["Country"].value_counts().plot(kind="bar")
```

```
[17]: <Axes: xlabel='Country'>
```



```
[18]: len(df)
```

```
[18]: 10
```

Figure 6

6.Filters

```
[12]: df["Country"].unique()
[12]: array(['India', 'Denmark', 'Australia', 'United Kingdom', 'Brazil',
        'Canada', 'Mexico'], dtype=object)
[15]: df[df["Country"]=="Canada"]
[15]:
```

	First Name	Last Name	Gender	Age	Country	Email
6	Zachary	Macdonald	male	36	Canada	zachary.macdonald@example.com
9	Emilie	Mitchell	female	62	Canada	emilie.mitchell@example.com

```
[16]: df[df["Age"] > 40]
[16]:
```

	First Name	Last Name	Gender	Age	Country	Email
0	Sayali	Patil	female	52	India	sayali.patil@example.com
1	Silas	Johansen	male	79	Denmark	silas.johansen@example.com
2	Alma	Nielsen	female	67	Denmark	alma.nielsen@example.com
3	Hilda	Phillips	female	45	Australia	hilda.phillips@example.com
4	Jordan	Nguyen	male	42	United Kingdom	jordan.nguyen@example.com
7	Gordon	Davis	male	67	Australia	gordon.davis@example.com
9	Emilie	Mitchell	female	62	Canada	emilie.mitchell@example.com

Figure 7

7.Streamlit run app.py

localhost:8501

Deploy

User Profile Insights

	First Name	Last Name	Gender	Age	Country	Email
0	Hrithik	Kamath	male	38	India	hrithik.kamath@example.com
1	Alejandro	Rubio	male	67	Spain	alejandro.rubio@example.com
2	Trinidad	Vega	female	73	Spain	trinidad.vega@example.com
3	Enora	Michel	female	72	France	enora.michel@example.com
4	Dragoljub	Isaković	male	25	Serbia	dragoljub.isakovic@example.com
5	Praneel	Kumari	male	66	India	praneel.kumari@example.com
6	Laurénio	Peixoto	male	68	Brazil	laurenio.peixoto@example.com
7	Daniela	Pérez	female	73	Spain	daniela.perez@example.com
8	Howard	Thomas	male	57	United Kingdom	howard.thomas@example.com
9	Elmer	Vikane	male	33	Norway	elmer.vikane@example.com

7.Conclusion:

-تم تحقيق اهداف المشروع:

١-التعامل مع API.

٢-استخدام Pandas.

٣-تحليل البيانات.

٤-عرض البيانات.

٥-تشغيل المشروع باستخدام Streamlit.